

Loan-Approval Prediction Classifier App User Interface (Build by Gradio)

Loan Approval Prediction App

Enter Loan Application Details to Predict Approval

Gender	Male
Marital Status	Married
Education	Postgraduate
Employment Status	Self Employed
Occupation Type	Freelancer
Residential Status	Rent
Region	Urban
Loan Purpose	Personal
Loan Type	Secured
CoApplicant	No
Loan History	No
Age	38
Dependent's Persons	5
Number of Served Loan	0
Interest Rate	11.5
Annual Income	80000
Monthly Expenses	60000
Total Existing Loan Amount	0
Outstanding Debt	0
Loan Amount Requested	200000
Loan_Months	60

Loan Status
Loan Approved

Flag

Clear

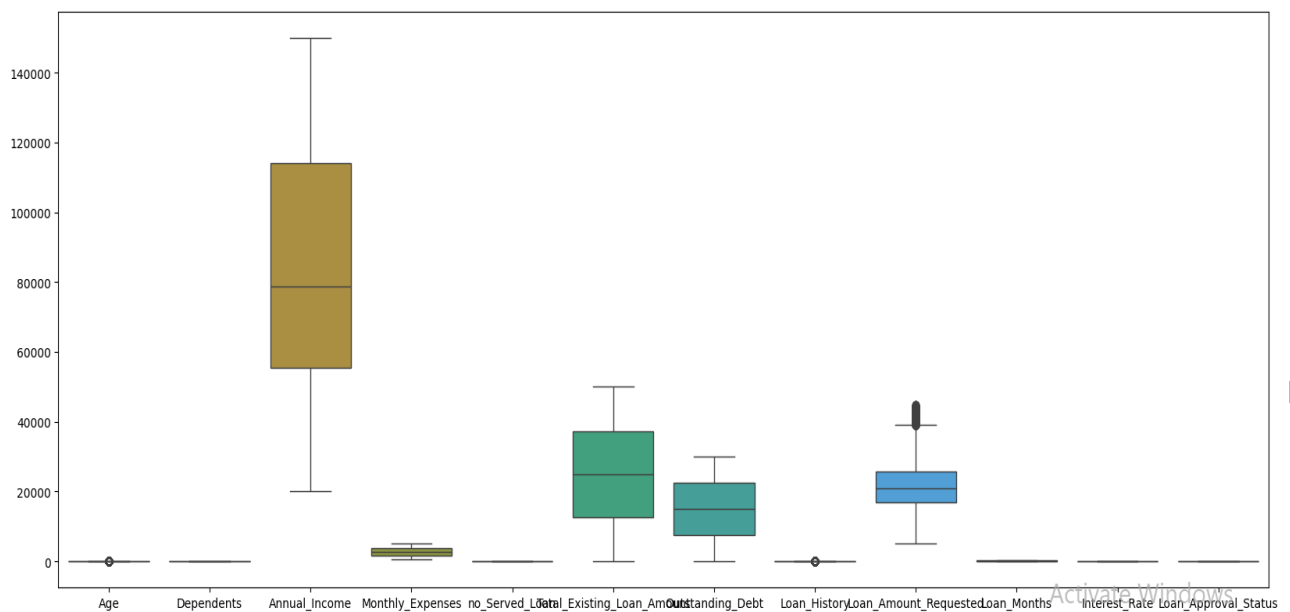
Submit

Checking Outliers By Plotbox Graph

Annual_Income, Applicant_ID, Existing_Loan_Amount, Outstanding_Debt, and Loan_Amount_Requested show the largest spread, indicating high variability in these features.

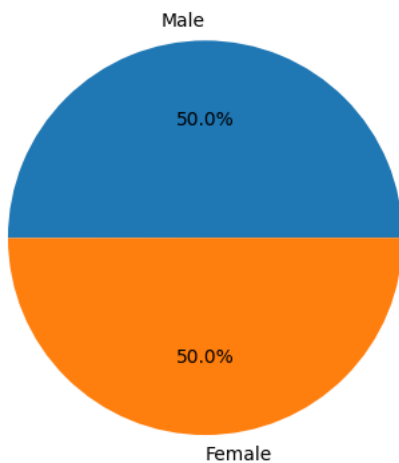
Loan_Amount_Requested contains several outliers, while most other variables show few or no noticeable outliers.

Binary/categorical features such as Loan_History, Existing_Loans, Loan_Term, Interest_Rate, Bank_Account_History, Transaction_Frequency, Default_Risk, and Loan_Approval_Status are compressed near zero because they are plotted on the same scale as large numeric features.



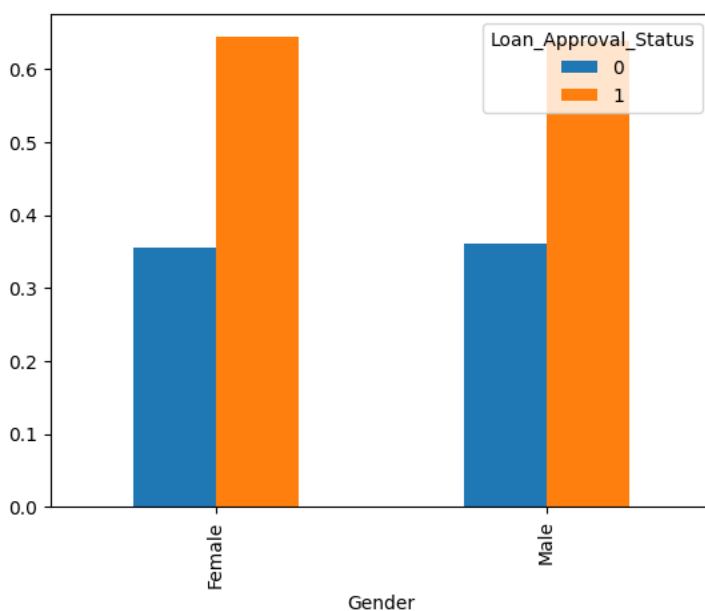
No Of People Who Took Loan By Gender

The dataset is perfectly balanced by gender, with 26,011 male applicants (50.0%) and 25,989 female applicants (50.0%). This balanced gender distribution helps reduce gender-related sampling bias, making the dataset suitable for fair model training.



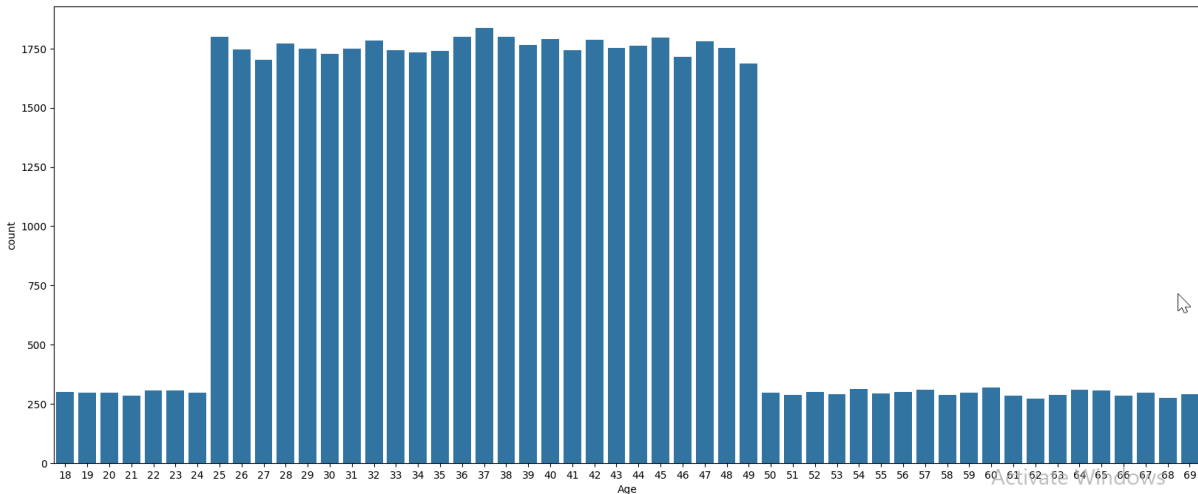
Observe loan status column as per Gender by bar grap

Gender has little to no impact on loan approval, as both males and females have nearly identical approval and rejection rates.



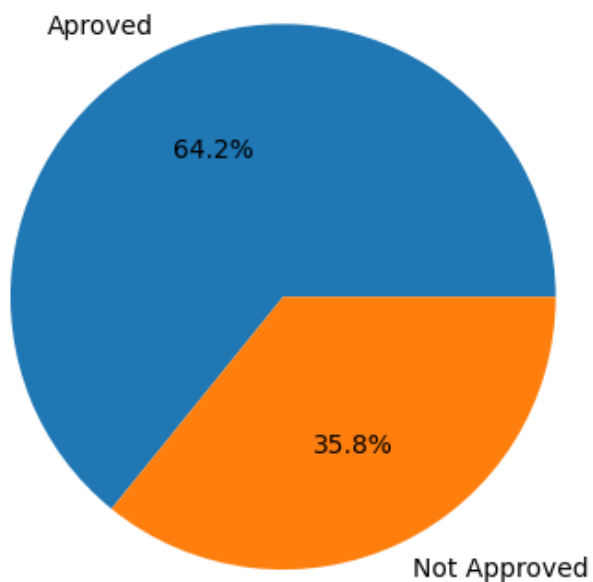
No Of People Who Took Loan By Age By Countplot

The age distribution is highest and relatively uniform between 25 and 49 years, indicating that most applicants belong to the working-age population. Fewer applicants are below 25 or above 50 years, showing that younger and older age groups are underrepresented in the dataset.



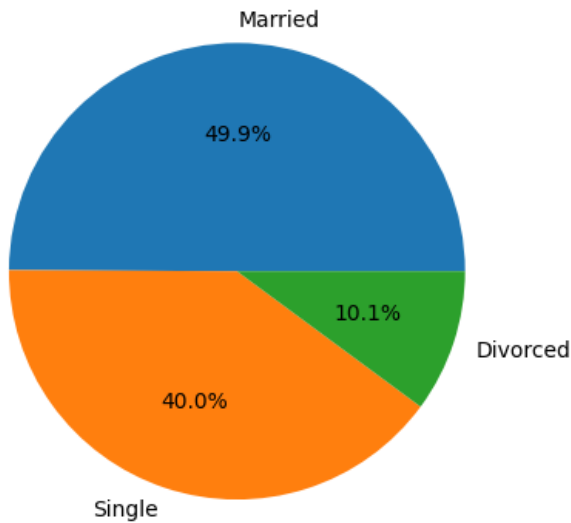
Observe Loan Status Column By Pie Graph

The dataset is moderately imbalanced, with 64.2% of loan applications approved and 35.8% not approved.



No Of People Who Took Loan By Marital Status By Pie Graph

Married applicants constitute the largest group (49.9%), followed by single applicants (40.0%), while divorced applicants represent the smallest proportion (10.1%) of the dataset.



Observe Loan Status Column As Per Marital_Status

Marital status has minimal impact on loan approval, as divorced, married, and single applicants all show nearly identical approval and rejection rates.

